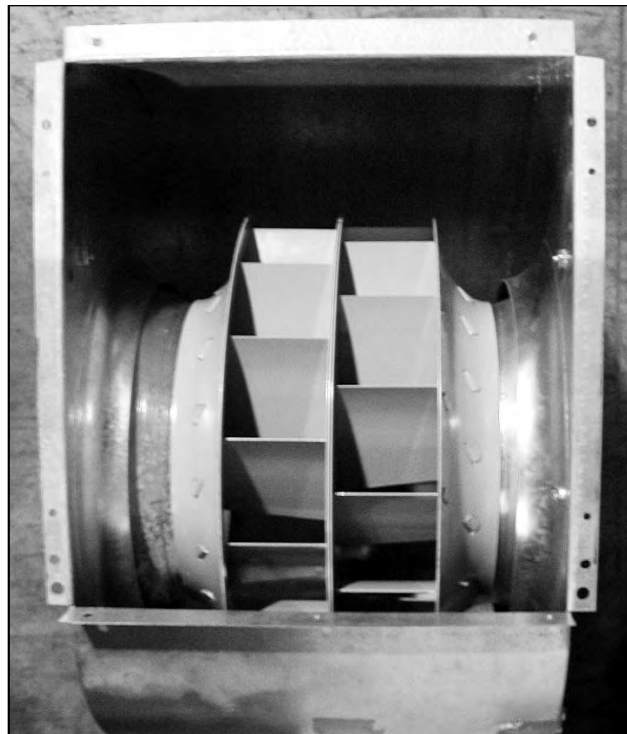


Reducing Air Handler Electricity Use: More Than Just a Better Motor

These days, manufacturers are able to produce relatively efficient appliances for people's homes. It is not uncommon to see A/C units with SEER 14, or furnaces with annual fuel utilization efficiency (AFUE) greater than 90%. Unfortunately, these numbers tell only part of the story. Air handlers in these systems are horribly inefficient, typically operating with efficiencies of 10%–15%. Several studies have looked at electric motor improvements (see "The Electric Side of Gas Furnaces," p. 24). That being said, it's time to look at the aerodynamic improvements of the fans in combination with motor improvements. The typical furnace fan suffers from the following problems:

- Clearances between fan blades and scroll housing are too large—a result of large tolerances in production (see photo on p. 9). Air just sloshes around the blade tips rather than moving into the ducts.
- The blades are generally stamped out of sheet steel and are not designed for optimal aerodynamic performance (see "Fanning a Revolution," *HE* July/Aug '03, p. 18).
- Forward-curved blades are generally less efficient than backward-curved blades (although they provide a relatively more constant flow over a wide pressure range).
- Blower inlets have sharp edges, which disrupt airflow into the fan.

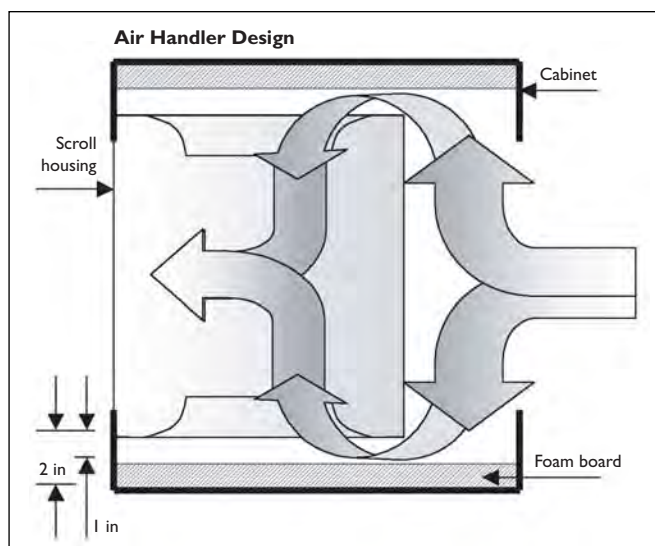


IAN WALKER

The prototype air handler features aerodynamically improved, backward-curved blades and a very small gap between the blower wheel and the housing.

Big Fan in a Little Cabinet

During fieldwork we've noticed that furnace cabinets have kept getting smaller, but the fans have stayed the same size. Convinced that this can't be good, we tested our prototype and production air handler in a restricted cabinet. We reduced the 2-inch clearance between the fan entry and the air handler cabinet to 1 inch (see Figure). The efficiency of the production air handler dropped from 15% to 11%. The prototype air handler was even more sensitive; it went from 21% to 8%. The tight clearance prevents smooth entry into the fan, negating the effect of a more aerodynamic design.



The addition of foam board restricts the fan inlet clearance, decreasing fan efficiency.

DEAN MINGEE



In contrast, the typical air handler has a gap that, at 1 inch, is eight times larger.

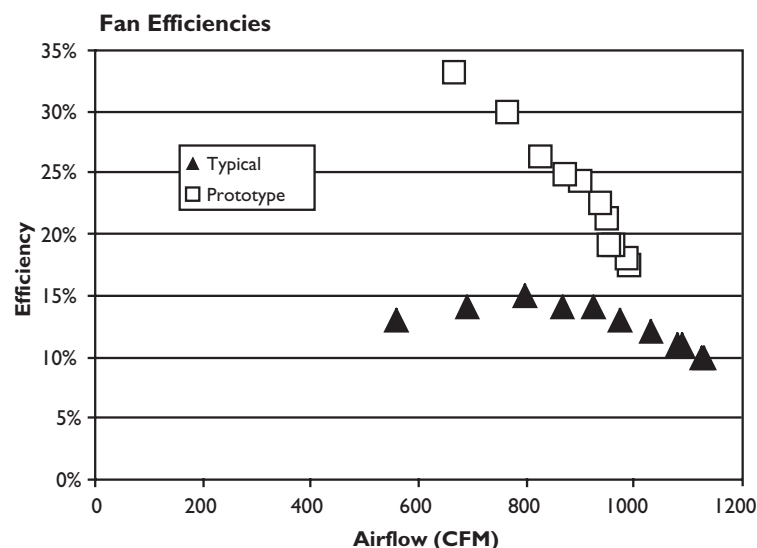


Figure 1. The efficiency of the prototype air handler increases at lower air flow.

And here are the improvements that got incorporated into a prototype air handler:

- Clearances between fan blades and housing were reduced to less than $\frac{1}{8}$ inch.
- Redesigned, backward-curved blades deliver better aerodynamic performance.
- Improved blades are combined with an electrically commutated motor (ECM) to maintain flow rates.
- Smooth, large-radius inlet cones create less noise and a better flow pattern entering the fan.

Recently, we tested this prototype (developed by General Electric in conjunction with DOE) in our full-scale duct system here at Lawrence Berkeley National Laboratory. The prototype uses the latest generation of ECM to control the fan speed, so our results include the increase in efficiency from having a good electric motor, as well as the increase in efficiency attributable to improvements in the fan. We measured the performance of both a typical production air handler and the new prototype over a range of system flow resistances (obtained by systematically closing registers).

The test results showed that the prototype outperformed the production air handler, being about twice as efficient overall, with an average efficiency of 21%, corresponding to about 3.5 CFM per watt (see Figure 1). The shapes of the two curves in Figure 1 are different because the standard air handler operates at a single speed and has an optimum efficiency near the operating point and decreasing efficiency at higher and lower flow rates. The prototype air handler operates at almost constant torque, and its speed is reduced at lower flow rates. This leads to increasing efficiency at lower air flows. The prototype's flow range was limited because its controller limited the speed range of the fan.

Most of the energy losses for the prototype are aerodynamic; the ECM operates at about 75% efficiency, and the fan has an aerodynamic efficiency that ranges from 15% to 45%, with an average of about 35%. (See "Big Fan in a Little Cabinet" for more on fan design and efficiency.)

One of the significant differences between the two air handlers was that the prototype becomes even more efficient at lower air flows. This makes these new devices even more appealing if you

are using them to ventilate a house, which requires much lower flows than are required during heating or cooling.

Switching to this prototype, with its ECM and aerodynamic fan design, can save 300 watts per house of excess heat produced by the typical motor and fan and put into the airstream. In turn, the A/C unit would use about 100 watts less by not having to take that heat out of the airstream (assuming a coefficient of performance of 3). In heating operation, the excess heat from the less efficient fan motor is not so important; but for gas or oil furnaces, changing to the more efficient ECM means that you are producing more heat from gas or oil, which is often cheaper than electricity.

The annual energy savings using this prototype fan would be about 500 kWh, or \$50 (assuming a rate of \$0.10 per kWh). Unfortunately, cost data are not yet available for the prototype, but the considerable savings indicate that more efficient air handlers will be cost-effective in most applications.



—Iain Walker and Dean Mingee

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